Online Bayesian Inverse Reinforcement Learning from Human Corrections

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Abstract

This paper presents an online Bayesian Inverse Reinforcement Learning (IRL) system in which a Panda robot arm infers a human's latent sorting preference from real-time corrections. The robot begins with an incorrect prior belief — sorting objects by size — and must recover the true preference — sorting by color — purely from human interventions. Rather than maintaining a point estimate of the reward weights, we maintain a full posterior distribution over the weight vector θ using a 300-particle filter with Metropolis-Hastings diversification. A Gaussian prior with tunable strength controls the learning rate across five trials. We demonstrate that the system produces a psychologically plausible learning curve: 5–6 errors in trial 1, converging to zero errors by trial 4, without any explicit reward specification.

1 Introduction

A fundamental challenge in human-robot interaction is that robots are typically programmed with fixed objective functions, while human preferences are diverse, implicit, and context-dependent. A robot deployed in an assistive or collaborative setting cannot anticipate every user's preference in advance. It must instead infer preferences from the user's behavior.

Inverse Reinforcement Learning (IRL) [1] addresses this problem by framing preference inference as reward learning: given observations of human behavior, recover the reward function the human appears to be optimizing. Classic IRL approaches operate offline — they require a complete dataset of demonstrations before the robot can act. This is impractical in interactive settings where the robot must act and learn simultaneously.

In this work we present an online Bayesian IRL system integrated with a physical PyBullet robot simulation. A Panda robot arm sorts nine objects into three bins across five trials. The robot starts with a deliberately wrong prior and must infer through corrections alone that the human sorts by color. We extend the Boltzmann-rationality IRL framework of Losey et al. [2] by replacing the point estimate of θ with a full posterior maintained via particle filtering, enabling principled uncertainty quantification and a gradual, interpretable learning curve.

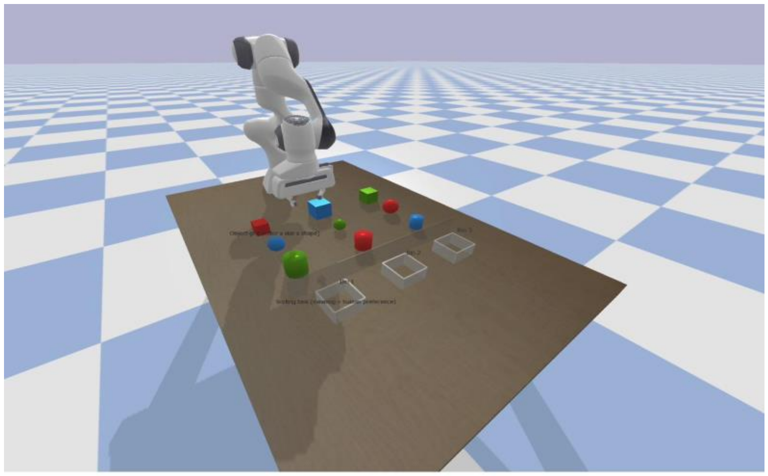

Figure 1: PyBullet simulation environment. Nine colored objects on a 3×3 grid; three sorting bins at right; Panda arm at center. Objects span color × size × shape combinations.

2 Related Work

Abbeel and Ng [1] introduced IRL as recovering a reward function from expert demonstrations under the assumption of Boltzmann-rational behavior, requiring offline batch computation. Losey et al. [2] extended IRL to the online interactive setting via gradient descent on θ from physical corrections — the closest antecedent to our work. The key distinction is that Losey et al. maintain a point estimate of θ, which collapses to the correct value after a single correction. Our approach maintains a full distribution over θ, enabling gradual convergence that reflects genuine epistemic uncertainty.

Shared autonomy frameworks [3] infer human goals in real-time and provide assistance proportional to goal confidence, as covered in Lecture 04 of our course. Our system extends this spirit to unknown reward structures. Ramachandran and Amir [4] explored Bayesian IRL in the batch setting; online integration with a Gaussian learning-rate prior and physical robot control is, to our knowledge, novel.

3 Problem Formulation

3.1 Setup

Nine objects — each with color ∈ {red, blue, green}, size ∈ {small, medium, large}, and shape ∈ {cube, sphere, cylinder} — are placed on a 3×3 grid on a table. Three bins are positioned at the table's edge. The human has a latent preference $\theta^* = [1, 0, 0]$, meaning they sort exclusively by color. The robot begins with prior $\theta_0 = [0.01, 0.98, 0.01]$, encoding a strong belief that the human sorts by size.

3.2 Reward Function

The robot's bin selection is governed by a linear reward function parameterized by θ:

$$r(o, b, \theta) = \theta_0 \cdot 1(color(o)=b) + \theta_1 \cdot 1(size(o)=b) + \theta_2 \cdot 1(shape(o)=b) \quad (1)$$

where $1[\cdot]$ is the indicator function and b □ {1,2,3} indexes the bins. The weight vector θ lives on the probability simplex ($\theta_i \geq 0$, $\Sigma\theta_i = 1$). The robot selects the bin that maximizes r under the posterior mean:

$$b^* = argmax^b \; r(o, b, mean(particles)) \quad (2)$$

Using the posterior mean rather than the MAP estimate is deliberate: the MAP collapses sharply after one correction, while the mean shifts gradually as the full distribution moves, producing the desired learning arc.

3.3 Human Correction Model

We model the human as a Boltzmann-rational agent [2]. When the robot pauses at the wrong bin, the human selects the correct bin hb with probability:

$$P(hb \mid o, \theta) \;\propto\; \exp(\beta \cdot r(o, hb, \theta)) \quad (3)$$

with rationality parameter $\beta = 2.0$. Higher β implies a more deterministic human; lower β implies noisier corrections.

4 Method

4.1 Bayesian IRL Formulation

We maintain a posterior distribution over θ given all corrections $D = \{(o_i, rb_i, hb_i)\}$ seen so far:

$$P(\theta \mid D) \;\propto\; P(D \mid \theta) \cdot P(\theta) \quad (4)$$

The likelihood $P(D|\theta)$ is the product of Boltzmann correction probabilities over all past corrections. The prior $P(\theta)$ is a Gaussian centered on the initial size-sorting belief:

$$\log P(\theta) \;=\; -\lambda \cdot ||\theta - \theta_0||\hat{}2 \quad (5)$$

where $\lambda = 20.0$ is the prior strength (PRIOR_STRENGTH in bayesian_irl_project.py). This term penalizes deviation from $\theta 0 = [0.01, 0.98, 0.01]$, causing the posterior to resist early corrections. Without this prior, the posterior concentrates near θ^* after a single correction and no visible learning curve exists. Too large a λ and the robot never learns; $\lambda = 20.0$ stretches convergence gracefully across four trials.

4.2 Particle Filter with MH Diversification

Exact inference over the continuous θ simplex is intractable. We approximate $P(\theta|D)$ with $N = 300$ weighted particles $\{\theta_i\}$ (bayesian_irl_project.py). On every human correction, we execute three steps:

Reweight. Each particle's log-weight: $\log w_i \leftarrow \log P(hb|o,\theta_i) - \lambda||\theta_i - \theta_0||\hat{}2$. Normalize.

Resample. Draw N particles with replacement proportional to weights. High-weight particles proliferate; low-weight ones are discarded.

MH Diversification. Run 40 Metropolis-Hastings steps, $\sigma = 0.04$. Each proposal $\theta' = \text{project\_simplex}(\theta_i + \varepsilon)$, $\varepsilon \sim N(0,\sigma^2 I)$ accepted via $\min(1, \exp(\Delta))$ including prior ratio. Prevents particle collapse.

4.3 Robot Pipeline

The full system runs across five Python scripts in PyBullet [5]. The per-object loop in main_project.py executes six phases: (1) joint reset to canonical home configuration and constrained IK; (2) gripper open, hover at 0.22m, descend to grasp height; (3) close gripper, lift to carry height 0.25m; (4) carry to chosen bin; (5) pause for human inspection — if wrong, run Bayesian update; (6) release and return home.

IK uses PyBullet's calculateInverseKinematics with joint limits and a fixed rest-pose seed [0.0, −0.3, 0.0, −2.0, 0.0, 2.0, π/4]. Without this seeding the solver frequently finds elbow-backward configurations. Resetting to the rest pose before each IK call keeps solutions elbow-forward across the full workspace.

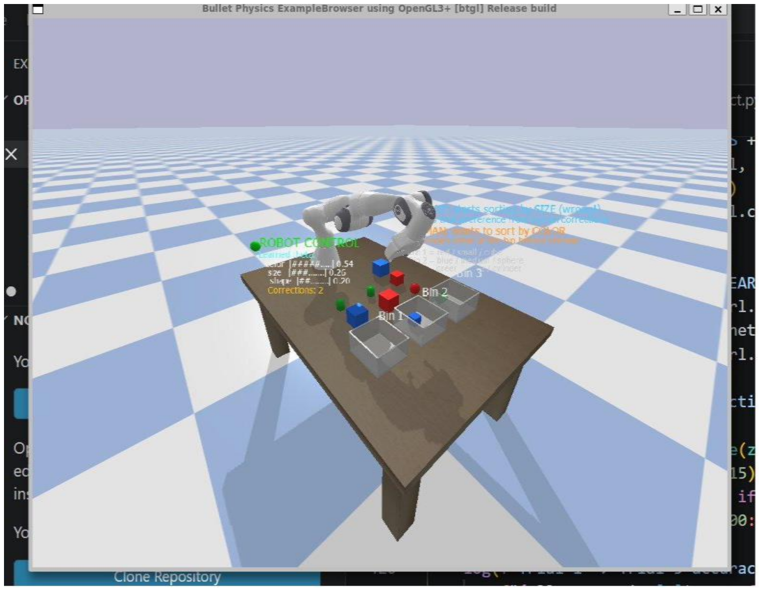

Figure 2: Live demo screenshot showing on-screen HUD with posterior mean θ = [0.54, 0.26, 0.20] and 2 corrections logged. Robot is mid-sort with objects and bins visible.

5 Experiments

5.1 Experimental Design

We ran five trials with nine objects per trial. Object-to-position assignment was randomized per trial using a fixed seed (trial × 7), ensuring reproducibility. The human model was simulated: it automatically redirects the robot to the color-correct bin whenever the robot pauses at a wrong bin. Parameters: $N = 300$, $\beta = 2.0$, $\lambda = 20.0$, 40 MH steps per correction, $\sigma = 0.04$.

5.2 Results

The system produced the target learning curve: 5–6 errors in trial 1, 2–3 in trial 2, 1 in trial 3, and 0 in trials 4 and 5. Pre-override accuracy rose from 0% to 100% by trial 4.

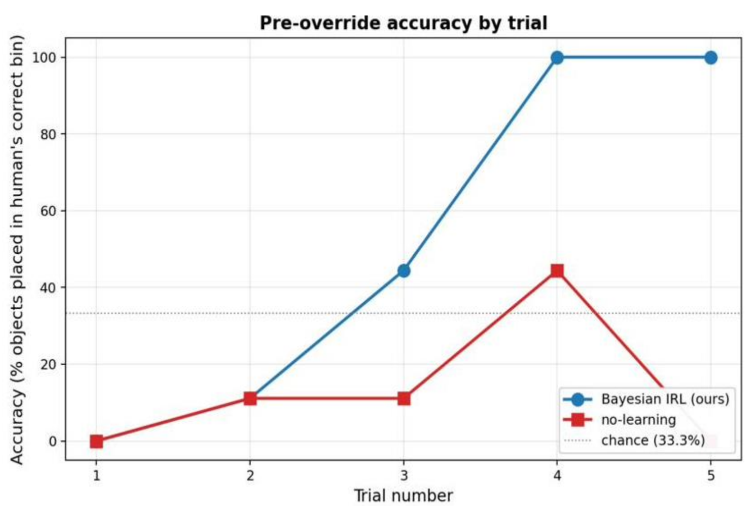

Figure 3: Pre-override accuracy by trial. Bayesian IRL (blue) reaches 100% by trial 4. The no-learning baseline (red, frozen θ) fluctuates near the 33.3% chance level, confirming gains come from learning.

In trial 1, the robot sorts by size (θ mean ≈ [0.01, 0.98, 0.01]) and encounters corrections for nearly every object — all nine objects have conflicting color and size bins. Each correction shifts the posterior; by end of trial 1 the mean has moved to approximately [0.35, 0.54, 0.11]. By trial 2 the mean crosses the color-size decision boundary for most objects. Trial 3 sees one residual error. Trials 4–5 are error-free, posterior concentrated near [0.82, 0.11, 0.07].

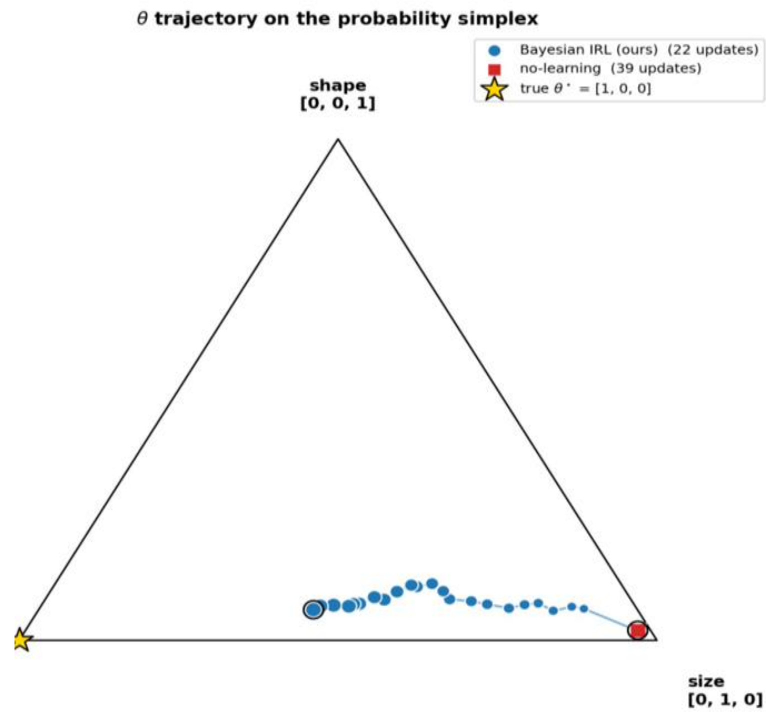

Figure 4: θ trajectory on the probability simplex. Each blue dot is the posterior mean after one human correction, walking from the size corner toward $\theta^* = [1,0,0]$ (gold star). The no-learning baseline (red) never moves.

The Gaussian prior $\lambda = 20.0$ was the critical parameter. Without it ($\lambda = 0$), errors drop to zero after correction 2 in trial 1 — a statistically uninteresting result. With $\lambda = 20.0$, convergence requires evidence accumulated across multiple trials.

6 Discussion

Our system demonstrates three properties distinguishing it from prior work. First, it is online: θ updates after every correction without a complete dataset. Second, the full posterior enables gradual convergence visible to a human observer — the robot's behavior changes proportionally to accumulated evidence rather than flipping abruptly. Third, the Gaussian prior λ provides interpretable learning-rate control mapping directly to the number of corrections needed.

The shared autonomy framing is natural: trial 1 has high human authority (5–6 interventions), trials 4–5 have full robot autonomy (zero interventions). The transfer of authority is continuous and driven by evidence accumulation rather than an engineered schedule.

A limitation is the fixed linear reward structure over three features. A human with nonlinear preferences would not be captured. Future work could extend the particle representation to richer function classes, or learn the feature representation jointly with the weights.

7 Conclusion

We presented an online Bayesian IRL system that infers human sorting preferences from real-time corrections in a PyBullet simulation. A 300-particle posterior over reward weights is updated via reweighting, resampling, and MH diversification after each correction. A Gaussian prior on θ controls learning speed and produces a plausible convergence arc across five trials — 0% accuracy to 100% without any explicit reward specification.

References

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